PROJECT SYNOPSIS

AI Resume Screening System

Synopsis

Submitted in Partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY in Computer Science & Engineering

Under supervision of [Supervisor Name] [Designation], Dept & School

Submitted By [Name1] ([Roll No1]) [Name2] ([Roll No2]) [Name3] ([Roll No3]) [Name4] ([Roll No4])

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CERTIFICATE

This is to certify that the project synopsis titled **"AI Resume Screening System"** submitted by **[Name1] ([Roll No.])**, **[Name2] ([Roll No.])**, **[Name3] ([Roll No.])**, and **[Name4] ([Roll No.])** under the supervision of **[Supervisor Name]** is a bona fide record of original work carried out at the School of Computer Science & Engineering, IILM University, Greater Noida.

Student(s) Signature Supervisor Signature

ABSTRACT

In the competitive landscape of modern recruitment, Human Resources (HR) departments face the daunting task of manually screening thousands of resumes for limited job openings. This manual process is time-consuming, prone to human bias, and often results in inefficiencies. The **AI Resume Screening System**

aims to revolutionize this process by automating the initial screening of candidates.

This project leverages advanced Machine Learning (ML) and Natural Language Processing (NLP) techniques to evaluate candidate resumes against specific job descriptions. The system extracts text from various resume formats (PDF, DOCX), preprocesses the data, and utilizes TF-IDF vectorization and Cosine Similarity to calculate a relevance score. Furthermore, it integrates a Large Language Model (LLM) to provide qualitative feedback on the candidate's strengths and weaknesses.

The expected outcome is a robust, user-friendly web application that significantly reduces the time-to-hire, ensures objective evaluation, and provides candidates with constructive feedback. The system features a unique "Brutalist" UI design for an engaging user experience.

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Chapter 1

Introduction

1. Introduction of Project

The **AI Resume Screening System** is a cutting-edge web application designed to assist HR professionals and recruiters in efficiently managing the high volume of job applications. By automating the preliminary screening process, the system allows recruiters to focus on the most promising candidates. The application uses Machine Learning algorithms to match resumes with job descriptions based on keyword similarity and semantic relevance. It provides a dual-scoring mechanism: a statistical match score derived from TF-IDF and Cosine Similarity, and a qualitative score generated by an AI model that analyzes the context of the resume.

2. Objective of the Project

The primary objectives of this project are: * **Automation:** To automate the tedious process of manual resume screening. * **Efficiency:** To reduce the time and effort required to shortlist candidates. * **Objectivity:** To minimize human bias in the initial screening phase. * **Feedback:** To provide candidates with instant, actionable feedback on their resumes. * **User Experience:** To create an engaging and modern user interface using "Brutalist" design principles. * **Accuracy:** To achieve high accuracy in matching resumes to job descriptions using advanced NLP techniques.

3. Scope of the Project

The scope of this project encompasses the development of a full-stack web application. * **Functional Scope:** * Resume parsing (PDF, DOCX). * Job description input. * Keyword matching and scoring. * AI-driven feedback generation. * Visual representation of scores and feedback. * **Boundaries:** The system is currently designed for text-based resumes and English language content. It does not currently handle image-based resumes without OCR (though OCR is planned). * **Impact:** * **Industry:** Streamlines the recruitment workflow for companies of all sizes. * **Society:** Promotes fair and merit-based hiring practices by reducing unconscious bias.

4. Literature Review

Resume screening has evolved from manual keyword searching to advanced AI-driven systems. * *Reference 1:* Early systems relied on simple boolean search and keyword counting, which often failed to capture context (e.g., "Java" the coffee vs. "Java" the language). * *Reference 2:* The advent of TF-IDF (Term Frequency-Inverse Document Frequency) allowed for better weighting of important terms, improving matching accuracy. * *Reference 3:* Recent advancements in Transformers (BERT, GPT) have enabled semantic search, where the system understands the *meaning* of skills and experiences, not just exact matches. This project builds upon these advancements by combining traditional statistical methods (TF-IDF) with modern LLM capabilities.

5. Work Proposal

5.1 Hardware and Software Requirement

Table 5.1: Hardware Requirements

Component	Minimum Requirement	Recommended
Processor	Intel Core i3 / AMD Ryzen 3	Intel Core i5 / AMD Ryzen 5 or higher
RAM	4 GB	8 GB or higher
Storage	10 GB free space	50 GB SSD
Internet	Standard Broadband	High-speed connection (for API calls)

Table 5.2: Software Requirements

Component	Specification
Operating System	Windows 10/11, macOS, or Linux
Frontend Framework	Next.js (React), Tailwind CSS
Backend Framework	FastAPI (Python)
Programming Languages	Python 3.9+, TypeScript
Database	Local File System / CSV (Prototype), PostgreSQL (Production)
IDE	Visual Studio Code
Version Control	Git & GitHub

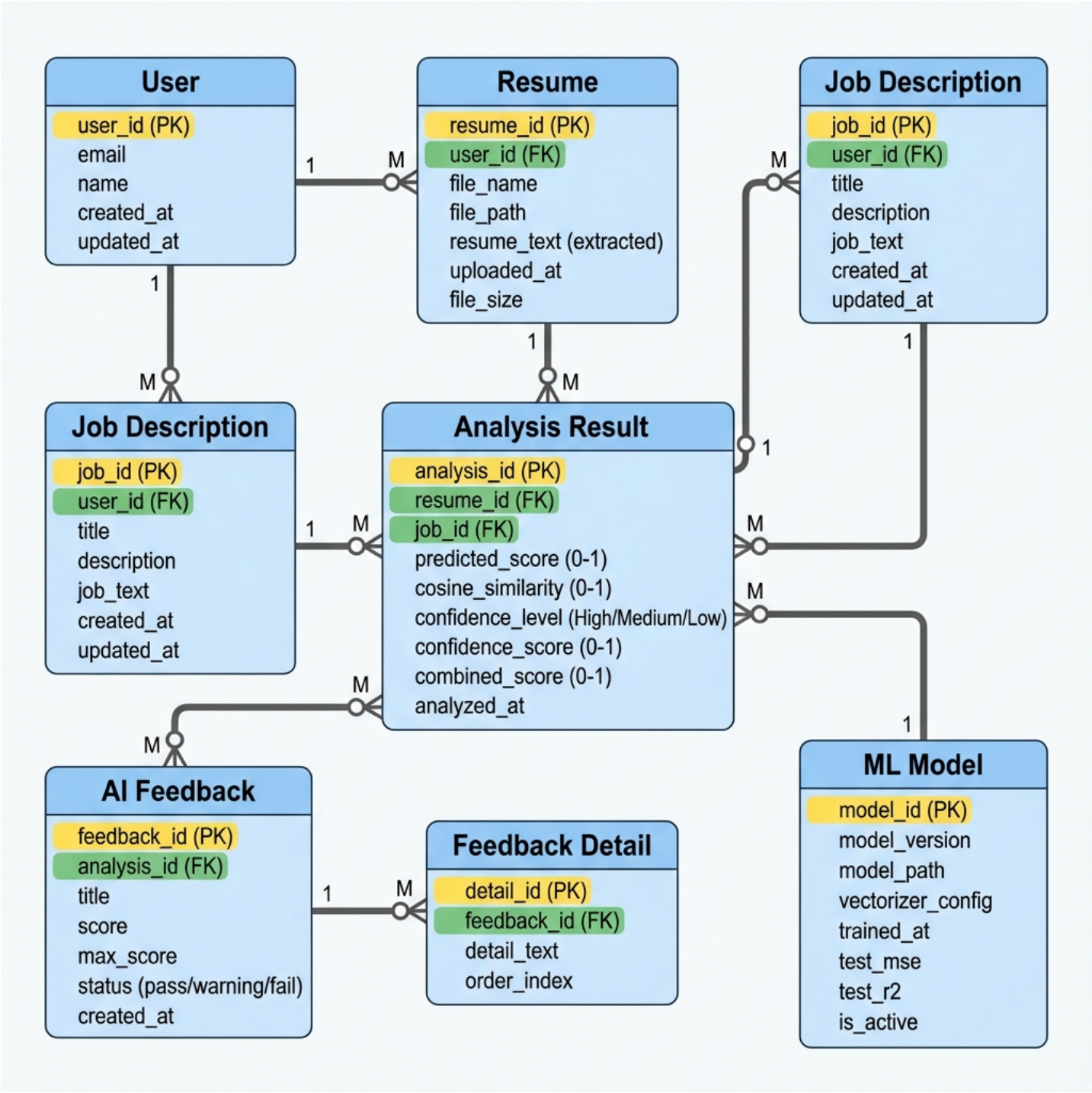
5.2 Database Requirement

For the current phase of the project, the system utilizes a file-based approach and in-memory processing for immediate analysis. * **Resume Data:** Stored temporarily for processing or in a CSV format

(`resume_data.csv`) for training the ML model. * **Model Artifacts:** Serialized Python objects (`.pkl` files) for the trained ML model. * **Future Database:** A relational database (PostgreSQL) or NoSQL database (MongoDB) will be required to store user profiles, historical application data, and detailed feedback logs.

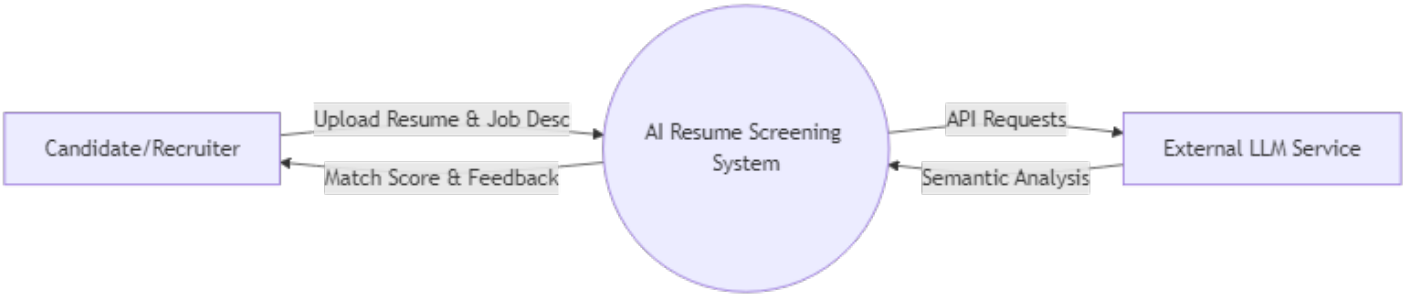
5.3 ER- Diagram

The Entity-Relationship (ER) diagram represents the data structure of the system.



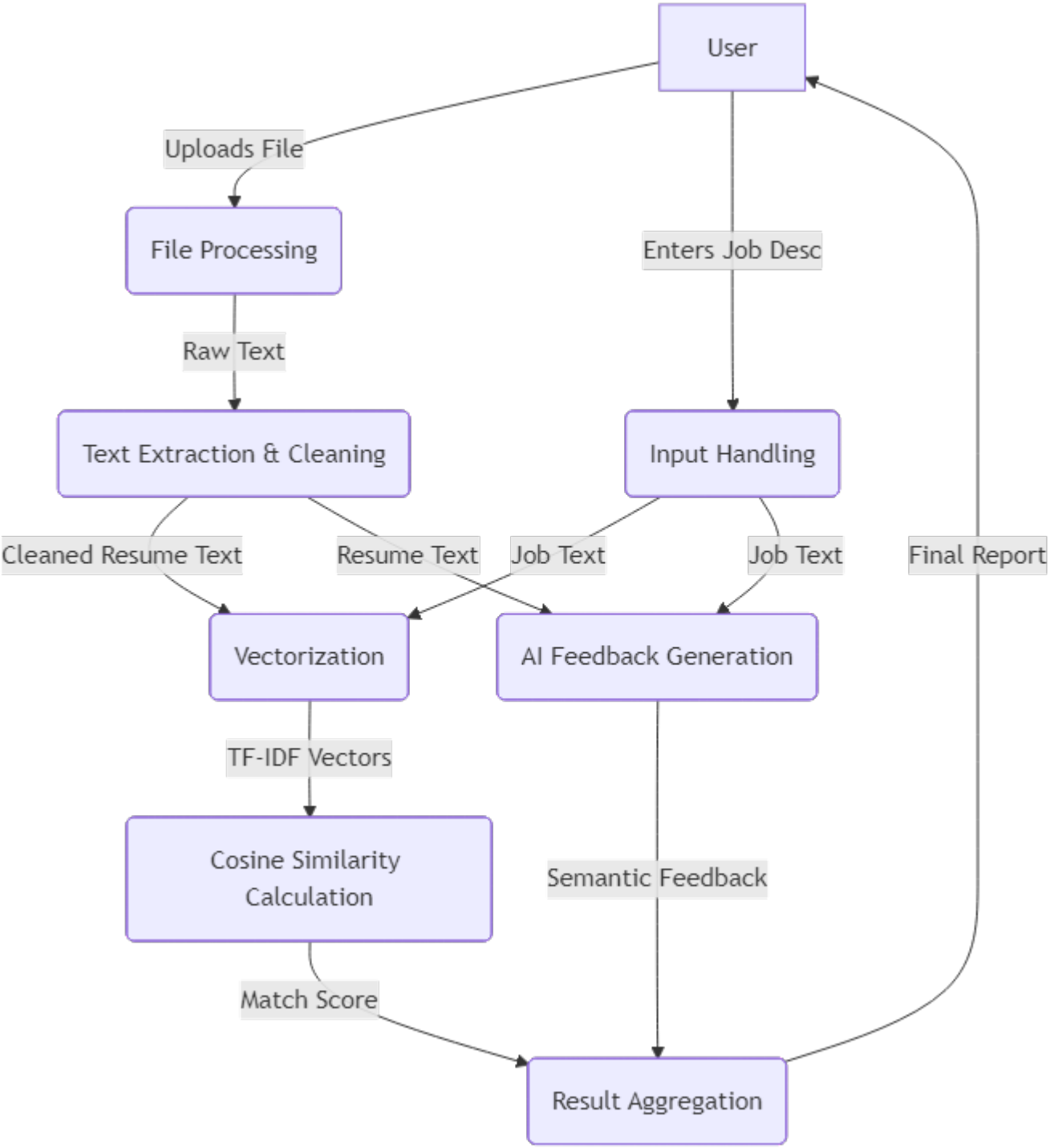
5.4 Level 0 DFD

The Level 0 Data Flow Diagram (Context Diagram) shows the system as a single process interacting with external entities.



5.5 Level 1 DFD

The Level 1 DFD breaks down the main system into sub-processes.



5.6 Methodology

The project follows the **Agile** development methodology, allowing for iterative improvements and flexibility. 1. **Requirement Gathering:** Analyzing the needs of HR professionals and candidates. 2. **System Design:** Creating the architecture, UI/UX designs (Brutalist style), and database schema. 3. **Implementation:** * **Frontend:** Developing the Next.js application with file upload and result visualization components. * **Backend:** Building the FastAPI service for text processing and ML inference. * **ML Model:** Training the TF-IDF model on a dataset of resumes to learn keyword importance. 4. **Testing:** Unit testing of individual components and integration testing of the full workflow. 5. **Deployment:** Hosting the application on a cloud platform (e.g., Vercel, Render).

6. Future Scope of the Project

- **Multi-Resume Ranking:** Extending the system to accept a zip file of resumes and rank them all against a single job description.
- **Skill Graphing:** Visualizing the candidate's skills vs. required skills using knowledge graphs.
- **Interview Question Generation:** Automatically generating technical interview questions based on the gaps identified in the resume.
- **Bias Detection:** Advanced algorithms to explicitly detect and flag potential biases in job descriptions or screening criteria.

7. Conclusion

The AI Resume Screening System effectively addresses the challenges of manual recruitment by providing a fast, objective, and feedback-rich screening tool. By combining statistical text analysis with modern AI capabilities, it offers a holistic view of a candidate's suitability. This project not only streamlines HR operations but also empowers candidates to improve their profiles, creating a more efficient and transparent hiring ecosystem.

8. References

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